Posterior Integration and Volatility Inputs in WTI Asian Options

Integración posterior y volatilidad en opciones asiáticas WTI

Abstract

Objective: We assess how much posterior integration changes WTI average-price option values relative to changing the volatility input. **Methods:** We combine a price-curvature expansion, 10 million simulated histories, an original futures-curve-based Asian-option study, and a separately identified extended conditional settlement-mark benchmark. **Results:** The original October-expiry sample has a mean absolute integrated-versus-plug-in price difference of $0.000745/bbl. On 2,381 date-ordered settlement-proxy observations, historical posterior integration has mean absolute pricing error 2.6187/bbl versus 0.1075/bbl using earlier option-implied volatility. In the distinct 97,341-observation settlement-mark benchmark, the corresponding errors are 0.8040 and 0.2184/bbl. A Bayesian market-implied-volatility model shows no reliable improvement over simple implied-volatility persistence. **Recommendation:** Evaluate posterior integration and volatility information separately. **Limitations:** Settlement proxies, few original valuation dates, model dependence, and contemporaneous parity anchors in the extended benchmark limit interpretation. **Originality:** We distinguish posterior point-price corrections, model-price dispersion, and volatility-information effects. **Conclusion:** Volatility inputs dominate observed pricing differences under the studied assumptions.

Keywords: Bayesian option pricing; Asian options; WTI crude oil; volatility uncertainty; implied volatility.

JEL classification: C11; C15; C58; G13; G17; Q40.

Resumen

Objetivo: Evaluamos cuánto modifica la integración posterior el valor de opciones asiáticas sobre WTI frente al cambio de fuente de volatilidad. **Métodos:** Combinamos una expansión basada en la curvatura del precio, 10 millones de historias simuladas, un estudio con la curva de futuros y una evaluación independiente de cotizaciones de liquidación condicionadas. **Resultados:** En las opciones con vencimiento en octubre, la diferencia absoluta media entre integración posterior y sustitución por la media posterior es de $0.000745 por barril. En 2,381 observaciones ordenadas cronológicamente, el error absoluto medio es 2.6187 con volatilidad histórica y 0.1075 con volatilidad implícita anterior. En otra muestra de 97,341 observaciones, los errores respectivos son 0.8040 y 0.2184. Un modelo bayesiano con volatilidad implícita previa no mejora de forma concluyente la persistencia simple. **Recomendación:** Separar la integración posterior de la información usada para estimar la volatilidad. **Limitaciones:** Las liquidaciones, pocas fechas y los supuestos de valuación restringen la interpretación. **Originalidad:** Distinguimos correcciones puntuales, dispersión posterior e información de volatilidad. **Conclusión:** La fuente de volatilidad domina las diferencias de error observadas bajo los supuestos analizados.

Palabras clave: valuación bayesiana de opciones; opciones asiáticas; petróleo WTI; incertidumbre de volatilidad; volatilidad implícita.

Clasificación JEL: C11; C15; C58; G13; G17; Q40.

1. Introduction

WTI Average Price Options (APOs) allow market participants to manage risk in an arithmetic average of first-nearby crude-oil futures settlements rather than a single delivery-date price. Their valuation depends on the option payoff, the futures information available at the valuation date, and the volatility assumption. Bayesian methods also allow uncertainty about volatility to enter conditional theoretical values. But more elaborate posterior computation need not improve a price if its underlying information is poorly matched to the derivative being valued.

Our distinction has three parts. First, posterior-integrated pricing (PI) averages conditional values over the volatility posterior, whereas posterior-mean plug-in pricing (PM) evaluates the conditional value at one posterior mean. Their difference concerns the *point price*; it does not equal the dispersion of the conditional model-price distribution. Second, the source of volatility information is an independent empirical choice. Third, a posterior distribution for model parameters and a predictive interval for actual settlement-mark errors answer different uncertainty questions and should not be presented interchangeably.

We examine those distinctions in a 10-million-history synthetic study and in two observational exercises, each reported with its own pricing assumptions and information cutoffs. The first uses individual Barchart APO histories, a CL futures-curve-based one-factor Asian option approximation, 310 October-expiry contract-date observations in August–September 2026, 2,381 date-ordered conditional repricing observations across 15 dates, and a 253-observation cross-instrument check using earlier LO vanilla-WTI implied volatilities. The second uses a larger Databento AAO settlement archive and a *different*, reduced-form lognormal terminal-average pricing approximation. The second exercise evaluates 97,341 actual-flagged settlement marks on 189 dates and includes prior-date market-IV Bayesian forecasting and retrospective pricing-residual interval calibration.

The central empirical finding is not that a particular Bayesian prior dominates. In the original analysis, historical-volatility posterior integration changes point prices little while earlier option-implied volatility substantially reduces the errors of the tested historical volatility inputs. The distinct longer-date AAO exercise confirms the importance of the volatility *information source* but shows no reliable gain from Bayesian market-IV integration over carrying forward the previous implied volatility. Its uncertainty-calibration comparison also reveals severe undercoverage in part of 2026. These results are conditional: some settlement observations may be derived from an exchange valuation model, and the extended analysis uses same-day *disjoint-strike* parity inputs. The two empirical cohorts cannot be pooled as if they came from one identical model.

Our contribution is therefore a transparent separation of three effects often grouped under the label of Bayesian option pricing: posterior curvature corrections, choice of volatility signals, and predictive uncertainty calibration. The paper explains the economic and measurement limits of each. The next sections describe the model, the two independent empirical designs, the results and their implications.

2. Literature review

Asian options depend on a price average rather than a single terminal price. Geometric-average approaches and conditioning approximations are long-established tools for computing arithmetic-average option values (Curran, 1994; Kemna & Vorst, 1990). REMEF has published related Asian-option work: Matías González et al. (2019) study a stochastic optimal consumption/investment framework with Asian-option valuation, whereas Ortiz Ramírez & Jiménez Preciado (2026) apply realized volatility and mean reversion to the CEMEXCPO setting. Neither addresses our narrower comparison of posterior integration with the choice of volatility information for WTI average-price option settlement marks.

Bayesian option pricing is also well established. Guidolin & Timmermann (2003) study learning and option-price dynamics, while Rombouts & Stentoft (2014) report a small effect of parameter uncertainty on pricing errors in their equity-option setting. More recently, Sagor et al. (2026) propagate Bayesian uncertainty into illustrative European-option prices, including a WTI example. Those results do not evaluate observed WTI arithmetic-average option marks. We do not claim that integrating a parameter posterior, or finding a small integration correction, is new. Our narrower contribution is the comparison with earlier option-implied volatility signals and a simple persistence benchmark for WTI average-price options.

WTI APOs provide a different test setting. Market-specific models may account for stochastic convenience yields, jumps or richer futures dynamics (Ewald et al., 2023; Schwartz, 1997; Shiraya & Takahashi, 2011). We hold the one-factor option-pricing benchmark fixed in the original analysis, so the influence of volatility information is not conflated with changes in the dynamics of the commodity futures curve. This is deliberately a controlled comparison, not a search for the most elaborate structural commodity-price model.

The distinction between historical and option-implied information is important. Gilder & Tsiaras (2020) study forecasts of *subsequent realized crude-oil volatility* extracted from option prices. We study errors in pricing WTI *average-price options* at later valuation dates. The risk-neutral option-implied dispersion need not equal a physical-return volatility estimate, especially if the latter uses a proxy for the wrong underlying random quantity. Consequently, a large historical-versus-implied pricing difference by itself neither identifies a variance risk premium nor demonstrates that Bayesian inference fails.

The REMEF studies examine Asian-option valuation with stochastic-control or mean-reverting volatility specifications; the crude-oil option study addresses forecasts of subsequent realized volatility. They are complementary but do not directly test our question. Here the pricing dynamics are held fixed while we separate the posterior-integration correction, the dispersion of conditional model values, and the contribution of previously available volatility inputs to out-of-sample APO pricing errors.

3. Methodology

3.1 Physical-measure inference and risk-neutral valuation

Historical log returns are modeled under the physical measure as

$$R_i \sim \mathcal{N}\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t, \sigma^2 \Delta t\right],$$

with baseline priors

$$\mu \sim \mathcal{N}(0,1), \qquad \sigma \sim \text{InvGamma}(2,0.1), \qquad \sigma > 0.$$

We sample the empirical posterior in $(\mu, \log\sigma)$ using random-walk Metropolis. For the large synthetic experiment, we integrate out μ analytically conditional on σ. This leaves a one-dimensional posterior calculation and avoids MCMC error in that experiment.

Pricing is carried out separately under the risk-neutral measure. For futures maturity u, the maintained benchmark is

$$dF_t^u = \sigma F_t^u \, dW_t^Q,$$

where u indexes futures maturity and the same Brownian motion W^Q drives all maturities. The common factor specifies perfect instantaneous correlation across futures maturities under the maintained model. Physical drift does not enter risk-neutral valuation. Holding these dynamics fixed allows direct comparisons of posterior integration and volatility inputs.

For an arithmetic-average option with fixing dates $t_1, \ldots, t_M$,

$$A_T = \frac{1}{M}\sum_{j=1}^{M} F_{t_j}^{(1)},$$

and the conditional call value is

$$C_t^Q(\sigma) = B(t,T)\,\mathbb{E}^Q[(A_T - K)^+ \mid \mathcal{F}_t; \sigma],$$

where $B(t,T)$ is the discount factor, K the strike, and $\mathcal{F}_t$ the valuation-date information set. Put values are defined analogously. In the arithmetic-average definition above, $F_{t_j}^{(1)}$ is the first-nearby WTI futures settlement on fixing date t_j under the applicable exchange contract rules (CME Group, 2021). At a valuation date within the averaging period, earlier fixings are known. We simulate only the remaining fixings, using the corresponding contracts in the observed CL futures curve as initial values.

We price the baseline historical-volatility model by Monte Carlo. For the large out-of-sample comparisons and checks using alternative constant volatilities, we use Curran's deterministic conditioning approximation (Curran, 1994). We check numerical accuracy separately with high-precision pseudo-random Monte Carlo and scrambled Sobol calculations.

3.2 Posterior integration and price uncertainty

Let $\mathcal{D}$ denote the historical return data, $C^Q(\sigma)$ the conditional risk-neutral option value, and $\bar{\sigma} = \mathbb{E}[\sigma \mid \mathcal{D}]$. We compare posterior-integrated (PI) and posterior-mean plug-in (PM) prices:

$$C_{PI} = \mathbb{E}[C^Q(\sigma) \mid \mathcal{D}], \qquad C_{PM} = C^Q(\bar{\sigma}).$$

A second-order expansion around $\bar{\sigma}$ gives

$$C_{PI} - C_{PM} \approx \frac{1}{2} C^{Q\prime\prime}(\bar{\sigma}) \mathrm{Var}(\sigma \mid \mathcal{D}).$$

The approximation links the PI–PM difference to posterior variance and price curvature. Its sign depends on local curvature, and it is most accurate when higher-order terms are small over the posterior support.

The posterior distribution also induces a distribution of theoretical option values,

$$C^Q(\sigma), \qquad \sigma \sim p(\sigma \mid \mathcal{D}).$$

Its dispersion is approximately

$$\mathrm{SD}(C^Q(\sigma) \mid \mathcal{D}) \approx |C^{Q\prime}(\bar{\sigma})| \sqrt{\mathrm{Var}(\sigma \mid \mathcal{D})}.$$

The two quantities have different local orders in posterior dispersion. The PI–PM difference depends on posterior variance, whereas the standard deviation of conditional model prices depends on the posterior standard deviation of volatility. Neither quantity measures future market-price uncertainty or removes pricing-model misspecification.

3.3 Synthetic experiment

The synthetic experiment is designed to vary posterior variance and price curvature in the second-order approximation. It contains 10 million simulated return histories, generated with root seed 2468 and distributed over 35 combinations of historical sample size (21 to 1,260 observations) and annual volatility (0.10 to 0.80). We evaluate 168 option states varying maturity, moneyness, and the fraction of the arithmetic average already fixed. For each history and option state, we calculate the PI–PM difference, posterior volatility variance, local price curvature, and the second-order approximation. The volatility posterior is evaluated on a 4,097-point grid; computations are batched without changing the model or pricing grid.

This design isolates the conditions under which posterior integration matters and tests the curvature–variance approximation independently of market pricing errors.

3.4 WTI Average Price Options and market data

The empirical sample is built from individual Barchart histories for WTI Average Price Options. The raw archive contains 213 unique contracts and 11,179 contract-date observations, with expiry months ranging from September 2026 to June 2029. We compare model prices with the Barchart end-of-day `Latest` field, treated as a settlement-price proxy. It has not been independently certified as an official final exchange settlement or executable transaction quote. Calls and puts are retained as available, and moneyness is recomputed by valuation date rather than assigned once at the contract level.

For every observation, we identify the fixings already known and the futures contracts used for the remaining fixings. We initialize those contracts with the futures prices observed on

the valuation date. This matters because the first-nearby contract can change during the averaging month.

Historical volatility is estimated from WTI futures returns available by the valuation date. The baseline uses the labelled continuous/front-month `CL=F` return series as the physical-measure proxy. A robustness check reconstructs a first-nearby return history contract by contract from CL settlements and excludes returns spanning contract switches.

3.5 Forward-in-time volatility comparisons

We evaluate volatility inputs through date-ordered conditional APO repricing errors. The archived implied-volatility estimation inputs have observation dates strictly preceding each target date; vendor first-publication times and historical revision vintages have not been independently certified. Repricing uses the target-date futures curve, realized fixings, and discount factor; the volatility input alone comes from earlier option dates. Historical-volatility estimates follow the specified return-data cutoff in the archived calculations. This does not establish when every underlying market record first became available. The sample comprises 2,381 contract-date observations across seven expiries and 15 target dates.

The baseline integrates conditional prices over a volatility posterior estimated from an expanding return sample available at the stated valuation cutoff. This is an error comparison conditional on that day's observed futures curve and fixing information, not a pre-opening market-price forecast. Historical alternatives use rolling windows of 63, 126, or 252 returns, an exponentially weighted moving average (EWMA) with a 63-business-day half-life, and a horizon-matched Gaussian GARCH(1,1) forecast. For GARCH, parameters are estimated from earlier returns; a valuation-date futures return updates conditional variance, which is then forecast over the remaining fixings. The updated GARCH predictor therefore uses valuation-date end-of-day underlying information, rather than solely information strictly before that date; it remains a conditional same-day repricing comparison. We map these forecasts to the constant annualized volatility that matches the remaining average's variance under the one-factor pricing model. Full comparisons are in Online Supplement Section S1 and Table S1.

We obtain contract-level APO implied volatilities by inverting the Curran pricing approximation. For each expiry, the previous-day implied-volatility smile is estimated on the latest available earlier date. A second specification fits a smile using all earlier dates, giving more weight to recent observations. Both depend on moneyness and option type. Neither uses the target-date APO settlement to estimate volatility.

To assess the value of smile fitting, we also carry forward each contract's most recent implied volatility. We reprice with the valuation-date curve, discount factor, and fixing schedule. This same-contract benchmark covers 2,366 observations.

Pricing performance is summarized by mean error, mean absolute error (MAE), and root mean squared error (RMSE). Dependence across contracts observed on the same valuation date is preserved with valuation-date cluster bootstrap resampling.

3.6 Cross-instrument validation

The within-APO comparison draws implied volatility from earlier observations of the same option family. We therefore add a cross-instrument check using standard monthly American WTI futures options (CME product LO) from Databento. We match LO option settlements to the underlying CL futures and invert prices using a Cox–Ross–Rubinstein futures-option tree.

For each APO target date, only LO observations from earlier dates are used. Implied-volatility smiles are fitted separately for the underlying CL futures contracts needed by the October-2026 APO fixing schedule. The resulting maturity-specific volatilities are mapped into the maintained scalar-volatility APO pricing model. We compare 253 APO contract-date observations over 10 target dates for which both external specifications can be evaluated within the moneyness range observed in earlier LO data. Online Supplement Section S4 and Table S4 report checks of the constant-volatility mapping and the number of steps in the Cox–Ross–Rubinstein tree.

This comparison tests whether the implied-volatility advantage transfers across WTI option families; it does not identify a variance risk premium.

3.7 Extended conditional AAO settlement-mark study

We assess whether the volatility-input ranking is also observed in an independently collected AAO settlement-statistics archive. This analysis does *not* extend the earlier Curran/CL-fixing panel. Its inputs, pricing function, and information set differ. The Databento AAO definitions and settlement statistics cover January 2020–September 2026; admissible evaluation dates run from January 2023 through September 2026. A fixed five-source-day sampling stride and common contract eligibility filters give 189 valuation dates, 1,689 eligible expiry-date groups, and 111,585 contract-date targets. Of these, 97,341 are flagged as actual settlements and 14,244 as theoretical. All headline error comparisons below use the 97,341 actual-flagged targets.

A same-date cross-sectional put–call parity regression on a reserved set of matched strikes supplies the risk-neutral expected terminal average and an effective discount factor. The reserved strikes are excluded from evaluation targets. The subsequent price is the reduced-form Black price of a *lognormal terminal arithmetic average*, not the Curran approximation applied to a simulated path of first-nearby CL fixes. This contemporaneous use of other AAO option marks makes the exercise conditional on same-date market information; it is *not* an independent fully ex-ante pricing forecast.

The historical-volatility Bayesian input is estimated from lagged changes in the parity-implied expected AAO average for the same expiry, and thus is not a direct history of CL returns or a risk-neutral volatility estimate. Posterior integration (PI) and posterior-mean plug-in (PM) use the same return-proxy posterior. Rolling and exponentially weighted proxies are evaluated on exactly the same option targets. The unchanged lagged-implied-volatility benchmark inverts earlier-date near-the-money AAO option marks under the *same* terminal-average pricing proxy; no target-date implied volatility is used as a forecasting feature. All six methods use common contract/date support.

A second exercise uses only *strictly earlier* near-the-money AAO implied volatilities to model same-expiry log-implied-volatility innovations. Bayesian normal-inverse-gamma updating yields a Student-t posterior predictive log-volatility distribution: `bayes_IV_PI` averages option prices over 256 predictive volatility draws; `bayes_IV_PM` prices at their mean. Both are compared with the unchanged lagged-IV price on identical options. The chronological comparison is reported separately for 2023–2024 development dates and 2025–September 2026; the latter dates had already been inspected during initial work and are not an untouched confirmatory holdout for every subsequent diagnostic.

A final *distinct* calibration exercise constructs 90% and 95% intervals for observed-minus-predicted *settlement-mark errors,* using either strictly historical date-balanced Bayesian normal-inverse-gamma predictive residuals or rolling empirical residual quantiles. It uses a 63-date maximum window, requires at least 15 earlier dates per put/call-side and maturity stratum, and is scored by empirical coverage, interval width, and the proper interval score. These are *not* posterior credible intervals for an Asian payoff, nor posterior price bands from the volatility parameter in the earlier October-expiry study. Date- and calendar-month-block resampling is used for paired comparisons, not contract-row independence.

3.8 Computational implementation

The computations are implemented in Python using NumPy, SciPy and pandas for inference, simulation and market-data processing, and Matplotlib for figures. A public source-only repository contains the pricing library, experiment scripts, a synthetic demonstration, numerical tests, and source-provenance records. The original vendor observations and internal market-dependent outputs are retained privately. The synthetic calculation integrates physical drift analytically conditional on volatility. For empirical validation, estimated volatility inputs follow the documented time cutoffs, and errors are resampled by valuation date to retain within-date dependence.

3.9 AI-assisted manuscript preparation

OpenAI ChatGPT assisted with research software development, literature organization, language editing, and drafting and revision of the manuscript. The author remains responsible for the methods, numerical calculations, independent checking and interpretation of the results.

4. Results and discussion

4.1 When does posterior integration matter?

Figure 1 shows how the PI–PM difference varies across the synthetic design. The largest differences occur with short estimation windows, high volatility, long maturities, few realized fixings, and pronounced price curvature. Across the 10 million synthetic histories, the maximum observed absolute difference is 0.1565 for a normalized $100 forward, and the largest within-cell 99th percentile is 0.1273. These extremes exceed the differences observed in the October-2026 sample. The sample maximum depends on the number of simulated histories; the within-cell percentile is less sensitive to rare simulated outcomes.

The second-order approximation tracks the simulated differences closely: the correlation between cell-level mean PI–PM differences and the curvature–variance term is 0.9995.

Through-origin slopes are also close to one in the most extreme settings. These results support posterior variance and price curvature as the principal local determinants of the difference.

The PI–PM difference generally declines as the share of realized fixings increases. The relationship is not strictly monotone: in Panel C, the 252-day case initially rises because partial fixing changes both the remaining volatility exposure and local price curvature. The difference nevertheless tends toward zero as the averaging period ends.

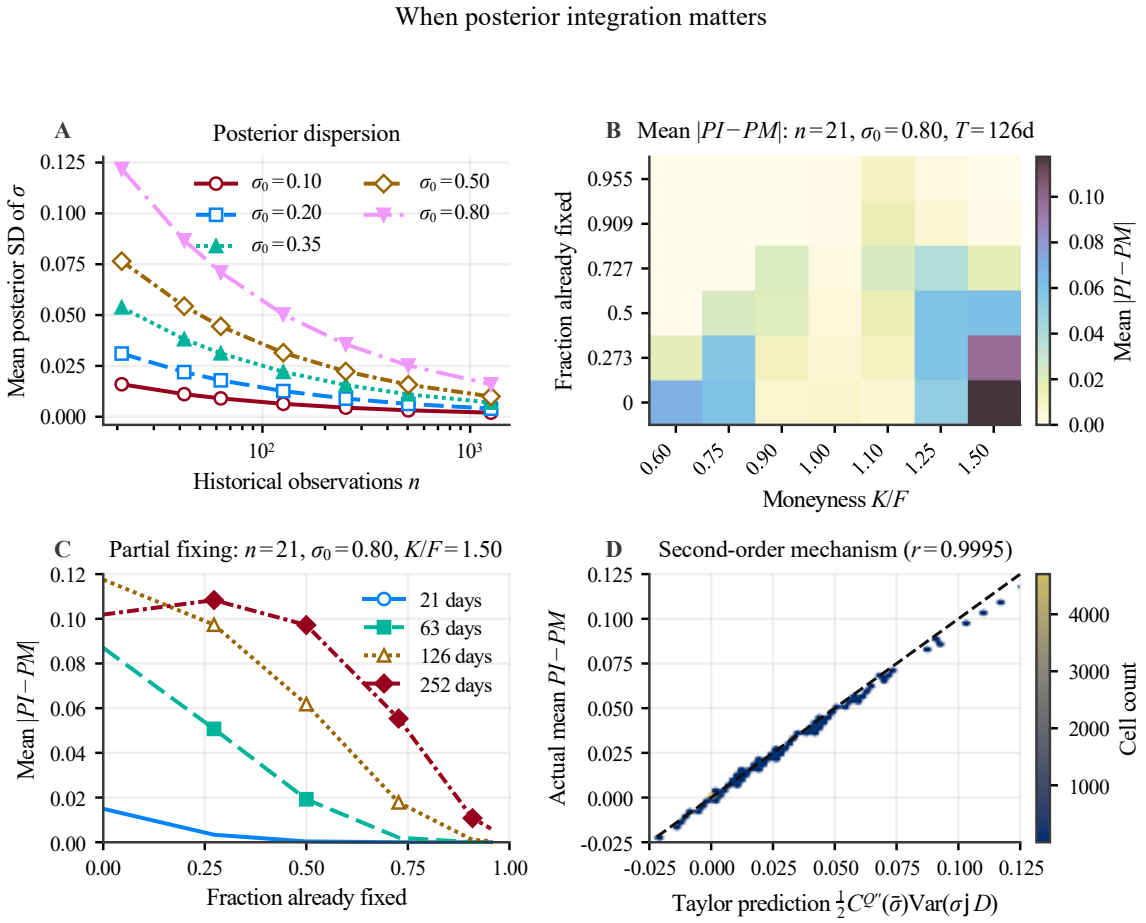


Figure 1. Posterior integration across 10 million synthetic return histories and WTI-like option states (root seed 2468). Panel A shows posterior dispersion by historical sample size. Panels B and C show the absolute PI–PM difference across moneyness and the realized-fixing fraction. Panel D compares mean PI–PM differences for 5,880 scenario cells with the second-order approximation in Equation (7); the cell-level correlation is 0.9995. The color bar shows the number of scenario cells per hexagonal bin.

Source: Authors' calculations from the recorded experiment outputs.

4.2 Posterior integration in the WTI sample

The sample of options expiring in October 2026 contains 310 contract-date observations across 12 valuation dates from August 25 to September 10, 2026. Table 1 shows nearly identical pricing errors for PI and PM (RMSE 0.5177 and 0.5183, respectively). Their mean absolute price difference is only 0.000745 dollars per barrel.

The central 95% posterior interval for conditional model prices has mean width 0.2733 dollars per barrel, approximately 367 times the mean absolute PI–PM difference. The nonzero dispersion of conditional model prices therefore coexists with an almost unchanged integrated point price. This interval is not an empirical forecast-error interval or evidence of executable economic value. Both methods underprice the reported October settlement proxies on average.

Table 1. WTI options with October 2026 expiry: posterior pricing and model-price uncertainty. Pricing error is model price minus settlement price. All price quantities and errors are in U.S. dollars per barrel.

Quantity	N	Value
PI RMSE	310	0.5177
PM plug-in RMSE	310	0.5183
Mean |PI − PM|	310	0.000745
Mean 95% posterior price-interval width	310	0.2733

Source: Authors' calculations from the recorded experiment outputs.

4.3 Historical versus option-implied volatility

Table 2 reports date-ordered conditional pricing errors on 2,381 contract-date observations over only 15 target dates. The historical PI benchmark has MAE 2.6187 and RMSE 3.4090, compared with 0.1075 and 0.1594 for the previous-day APO implied-volatility smile. The expanding earlier-date smile performs similarly. Both specifications reduce observed settlement-proxy errors for all seven expiries in the saved date-ordered design, with exploratory valuation-date cluster bootstrap intervals for the error differences below zero in each expiry (Figure 2).

Table 2. Historical-volatility and prior option-implied volatility benchmarks. Panel A uses the same 2,381 out-of-sample contract-date observations over 15 target dates. Panel B restricts the comparison to observations for which the same contract has a strictly prior implied volatility. Historical PI is the canonical Monte Carlo posterior-integrated price; scalar historical, GARCH, and implied-volatility inputs are repriced under the same Curran approximation. MAE and RMSE are in U.S. dollars per barrel.

Panel A: Main out-of-sample comparison

Volatility input	N	Dates	MAE	RMSE
Historical PI	2,381	15	2.6187	3.4090
Rolling 63 returns	2,381	15	4.5201	5.6465
Horizon-matched GARCH(1,1)	2,381	15	2.6730	3.4441
Previous-day APO implied-volatility smile	2,381	15	0.1075	0.1594

Panel B: same-contract lagged-IV benchmark

Volatility input	N	Dates	MAE	RMSE
Historical PI	2,366	15	2.6279	3.4173
Same-contract lagged IV	2,366	15	0.0992	0.1403
Previous-day fitted APO smile	2,366	15	0.1074	0.1594

Source: Authors' calculations from the recorded experiment outputs.

Rolling-window, EWMA, and horizon-matched GARCH(1,1) benchmarks do not close the pricing-error gap (Online Supplement Section S1 and Table S1). In particular, GARCH gives MAE 2.6730 and RMSE 3.4441 on the same observations.

Panel B isolates 2,366 observations with an earlier implied volatility for the same contract. Carrying that volatility forward yields lower pooled errors than the fitted previous-day smile (MAE 0.0992 versus 0.1074). This same-support negative result shows that the advantage over historical volatility does not require fitting an IV smile. Its 15 target dates do not support a general claim that persistence always outperforms a fitted smile.

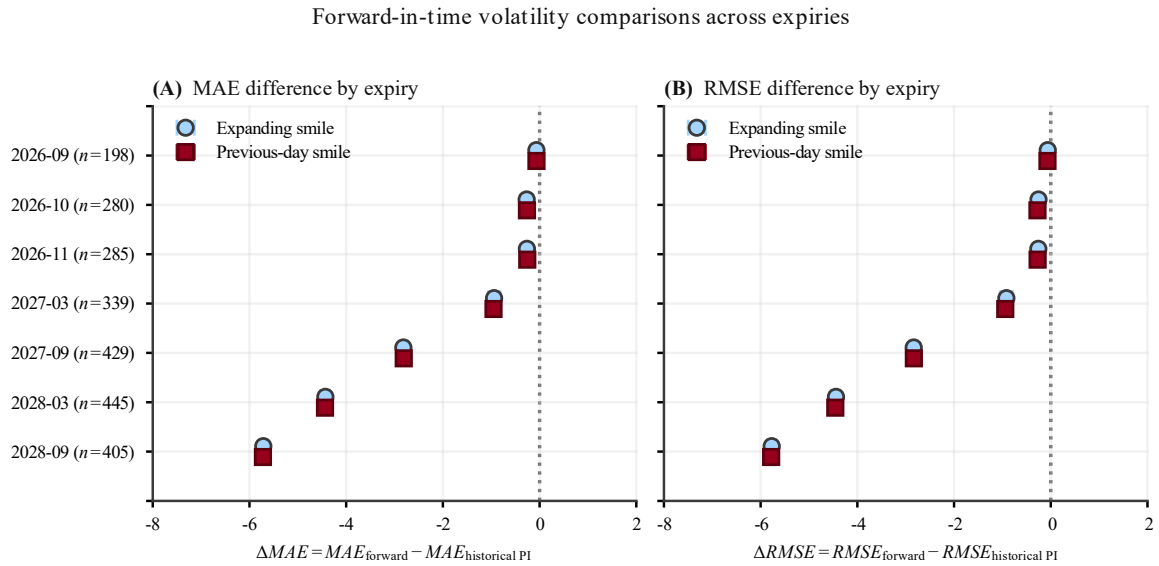


Figure 2. Out-of-sample pricing-error differences by APO expiry. Points compare the earlier-date APO implied-volatility specifications with the historical posterior-integrated benchmark; negative values indicate lower error for the implied-volatility specification. Horizontal bars are 95% percentile intervals from valuation-date cluster bootstrap resampling. All implied-volatility inputs have archived observation dates earlier than each target date; vendor publication times and historical record vintages are not independently verified.

Source: Authors' calculations from the recorded experiment outputs.

4.4 Cross-instrument validation using vanilla-WTI options

The cross-instrument comparison replaces APO-implied volatility with estimates from standard monthly WTI futures options. Table 3 uses 253 October-expiry APO contract-date observations across 10 valuation dates. Historical-volatility posterior integration has MAE 0.4464 and RMSE 0.5562. A previous-day vanilla-WTI implied-volatility specification based on LO options lowers these errors to 0.1559 and 0.2599. The corresponding previous-day APO implied-volatility smile has MAE 0.1676 and RMSE 0.2686.

Table 3. Cross-instrument validation on 253 October-expiry APO contract-dates within observed prior LO moneyness support. Historical PI is the canonical Monte Carlo posterior-integrated price; the vanilla-WTI and APO implied-volatility inputs are repriced with the deterministic Curran approximation. All implied-volatility inputs use only earlier-date option information. MAE and RMSE are in U.S. dollars per barrel.

Volatility input	MAE	RMSE
Historical PI	0.4464	0.5562
Previous-day vanilla-WTI IV	0.1559	0.2599
Previous-day APO IV	0.1676	0.2686

Source: Authors' calculations from the recorded experiment outputs.

Valuation-date cluster bootstrap results support the improvement over historical volatility. Differences between LO-implied and APO-implied specifications are small and do not establish consistent superiority by MAE and RMSE. The cross-instrument test instead shows that the advantage persists when implied volatility comes from another WTI option family.

4.5 Independent Databento settlement-mark results

The separate Databento analysis provides a longer-date check of the *volatility-input ranking*, not an additional sample from the earlier path-dependent Curran design. Table 4 reports errors for identical actual-flagged targets. The previous-date AAO implied-volatility benchmark gives an MAE of $0.2184 per barrel, versus $0.8040 for posterior integration of the historical parity-forward-change volatility proxy. Both estimates are evaluated within the same reduced-form terminal-average approximation. Posterior integration itself changes historical proxy pricing little: the PM MAE is $0.8068.

Table 4. Independent Databento AAO settlement-mark exercise on 97,341 common actual-flagged contract-date targets over 189 valuation dates. Pricing errors are in dollars per barrel. This is conditional reduced-form pricing with same-date disjoint parity anchors, not the earlier Curran/CL fixing-path study.

Volatility input and pricing method	MAE	RMSE
Earlier AAO option-implied volatility	0.2184	0.2979
Historical proxy, posterior integration	0.8040	1.1372
Historical proxy, posterior-mean plug-in	0.8068	1.1401

Source: Authors' calculations from the recorded experiment outputs.

The observed discrepancy is consistent with the mismatch between physical-style changes in a *conditional expected average* and the option-market-implied dispersion of the *terminal average*. A retrospective near-the-money inversion on the evaluation date gives mean implied volatility 30.97% across 1,689 expiry-date groups, compared with a Bayesian plug-in price-equivalent volatility of 26.68% and lagged AAO option-implied volatility of 30.88%. The target-date inversion is used *only to diagnose* pricing errors after scoring, never as an input to forecasts. These numbers do not identify a volatility risk premium.

Table 5 isolates the more demanding question: does a Bayesian market-volatility forecast outperform a simple persistence benchmark when both use the same prior option-market information? On 47,718 actual-flagged target observations across 88 dates in 2025–September 2026, it does not. The unchanged lagged-IV method has slightly lower MAE and RMSE; price integration is almost indistinguishable from the matched Bayesian plug-in price. The difference in equal-date daily MAE between unchanged lagged IV and Bayesian market-IV PI is -$0.00100/bbl, with a 95% calendar-month-block bootstrap interval of [-$0.00331,$0.00082] for the Bayesian improvement. The interval crosses zero. These small gaps should not be presented as economically significant advantages.

Table 5. Matched actual-flagged AAO pricing, 2025–September 2026; 47,718 contract-date targets and 88 valuation dates. Bayesian market-IV variants use strictly earlier option-implied volatility.

Method	MAE	RMSE
Unchanged earlier-date AAO implied volatility	0.2089	0.3014
Bayesian market-IV posterior-mean plug-in	0.2098	0.3023
Bayesian market-IV posterior integration	0.2098	0.3024

Source: Authors' calculations from the recorded experiment outputs.

The independent settlement-mark *residual-calibration* exercise uses 89,696 common actual-flagged targets on 174 evaluation dates after historical warm-up. In the inspected 2025–September 2026 subset, the Bayesian residual model, attached to the Bayesian market-IV PI point forecast, attains 86.43% coverage for nominal 90% intervals and 90.11% coverage for nominal 95% intervals. Corresponding date-balanced rolling empirical quantile coverages are 84.72% and 90.66%. Proper-score month-block comparisons do not establish a consistent Bayesian advantage. The model experiences substantial nonstationarity: nominal 90% Bayesian intervals cover only 59.0% and 56.9% of targets in March and April 2026, respectively. These figures concern *settlement-mark error intervals*; they do not contradict the earlier much wider posterior *conditional model-price* bands, which target a different random quantity.

4.6 Robustness, interpretation and limitations

Alternative physical-return assumptions do not materially change the comparison. A standardized Student-t model leaves the PI–PM difference small relative to historical-volatility pricing errors. Reconstructing first-nearby CL returns contract by contract gives a similar result: on the 2,381 out-of-sample observations, historical PI has MAE 2.6468 and RMSE 3.4451, compared with 2.6187 and 3.4090 using the continuous/front-month return proxy. Further historical-volatility benchmarks appear in Online Supplement Section S1.

The small empirical PI–PM difference is not attributable to numerical noise. High-precision Monte Carlo, independently scrambled Sobol simulation, and Curran conditioning agree on the difference to a precision well below its observed magnitude. Simulation-based calibration provides a separate check on the posterior implementation. The supplement also reports a posterior-RMS plug-in sensitivity (Section S2), alternative mappings from LO-implied volatilities to one APO volatility, and Cox–Ross–Rubinstein tree-refinement checks (Section S4). These numerical checks do not establish an exact arbitrage-free representation of the volatility surface.

The market evidence has important limits. In the original 2,381-row date-ordered comparison, 2,319 option-date rows have zero recorded trading volume. The historical-versus-implied ranking remains under positive-volume and open-interest restrictions, but these subsets are small and some expiry-level samples are particularly limited (Online Supplement Section S3). An exchange settlement proxy should not be interpreted as an executable transaction price. Moreover, the main pricing comparison contains only 15 target dates and the external validation only 10. Valuation-date cluster resampling preserves dependence among contracts observed together, but cannot eliminate serial dependence across adjacent dates.

Finally, the one-factor constant-volatility model excludes richer futures-curve dynamics, stochastic volatility, jumps, and convenience yields. Its implied volatilities are model-equivalent pricing inputs, not uniquely identified risk-neutral diffusion parameters or direct estimates of a variance risk premium. The out-of-sample results establish a difference in pricing performance for the tested inputs and model, not a structural physical-versus-risk-neutral volatility decomposition.

The newer Databento check adds dates but does not remove its measurement limitations. Forward averages and effective discount factors are inferred from other *same-date* option strikes; the historical volatility proxy is estimated from earlier parity-implied forward-average changes, not first-nearby CL realized returns; and the lognormal terminal-average approximation does not price the actual partly fixed Asian path. Original Barchart end-of-day `Latest` fields are settlement proxies rather than independently certified final exchange transaction prices. CME Clearing switched AAO end-of-day settlement valuations to its Curran model effective May 18, 2020 (CME Clearing, 2020). Thus, agreement with end-of-day marks can partly capture persistence or replication of exchange model values, rather than new information about tradeable option prices.

A very small price correction need not have economic value. For instance, the older October-expiry mean absolute PI–PM difference is about $0.000745/bbl, well below the standard $0.01/bbl quotation increment. For the extended conditional holdout, the tiny differences between the market-IV Bayesian models and unchanged lagged IV are not robustly distinct in the paired monthly resampling analysis. In addition, early-2026 settlement-mark residual-interval undercoverage warns against treating a well-fitted retrospective uncertainty distribution as a stable predictive probability model across regimes.

The older Barchart/LO evaluation and the newer Databento benchmark have different economic observables, providers, pricing engines, maturity support and information sets. They are *separate analyses*, not pooled observations for one common estimator. None directly validates executable hedge profits or the implied-volatility risk premium. Original vendor first-publication times, historical revision vintages, Barchart redistribution rights, trade/quote validation, and independent end-to-end historical input replay remain open. By contrast, stored-row arithmetic for the 2,381 fitted-smile and 2,366 same-contract carry comparisons has been independently reconciled, including all 4,762 fitted IV forecasts and archived prior-date relationships. These numerical checks do not prove when the vendor data first became available.

Compared with the related Asian-option valuation studies, these results do not establish that constant volatility is the best structural pricing model. They show how posterior integration and the choice of volatility input contribute differently to the pricing errors of one fixed benchmark.

5. Conclusions and recommendations

We examine two distinct effects in WTI average-price-option valuation: changing a volatility estimate and averaging model prices over uncertainty about that estimate. Posterior integration affects the conditional *point price* through volatility posterior dispersion and option-price curvature; the dispersion of conditional prices induced by the posterior is a different object. Large synthetic simulations illustrate the mechanism. In the original October-2026 *expiry* sample, the posterior integrated and plug-in values are nearly identical despite appreciably wider posterior distributions of conditional model prices.

The original date-ordered Barchart/LO comparison finds much smaller conditional settlement-proxy pricing errors when the volatility input comes from earlier option prices rather than the tested historical-return models. On matched observations, carrying forward

the same contract's last implied volatility also outperforms a fitted previous-day smile on pooled MAE, without establishing universal superiority. An independent longer Databento AAO settlement-mark exercise also favors earlier AAO implied volatility, although its pricing function and information set differ. Within that reduced-form exercise, introducing a Bayesian model for earlier implied-volatility innovations does not reliably improve on carrying forward the latest implied volatility. Additional Bayesian modeling of historical settlement-mark residuals does not maintain nominal interval coverage during early 2026.

The practical recommendation is to compare Bayesian posterior-integrated option values with a matched plug-in price, benchmark the volatility input against an honestly lagged option-implied alternative, and report *model-price posterior dispersion* separately from empirical forecast-error coverage. These comparisons should be interpreted relative to the specific market observations and pricing assumptions, not as a general ranking of stochastic-volatility models.

The evidence does not establish that the reduced-form terminal-average approximation or the maintained one-factor futures dynamics are structurally correct. End-of-day settlement fields can be model-generated, and the older independent-date market comparisons remain small. Original option-mark inputs have only date-level timing, and most of the original comparison rows report zero trading volume. Additional work with independently time-stamped CL fixing histories, market quotes and carefully designed hedging tests is needed before attributing pricing differences to tradable market inefficiencies or to a variance risk premium.

For valuation practice, the results support benchmarking historical volatility against earlier option-implied information and reporting posterior model-price dispersion separately from the PI–PM correction. Future work should expand the number of valuation dates and liquid contracts and examine stochastic-volatility and multifactor alternatives.

Data availability

The primary empirical study draws on Barchart WTI Average Price Option end-of-day fields (which can serve as settlement proxies), CL futures contract histories, and Databento standard WTI/LO option data. A distinct longer-date conditional benchmark uses Databento AAO option definitions and settlement statistics; its option forward-average and discount inputs are estimated from other same-date AAO strikes by put–call parity. These inputs are not equivalent to direct contemporaneous CL fixing histories. Source market records require vendor permission before redistribution; first-publication timestamps and historical data revisions are not independently certified. Replication requires independently licensed observations, source-vintage verification and the computational workflow specified in the supporting research records.

Code and results availability

The source-only companion software contains pricing and Bayesian-inference code, synthetic demonstrations, numerical tests, and scripts for the empirical analyses. Its identifying repository location is omitted here to preserve double-blind review and may be provided confidentially to the editor. Re-executing empirical experiments requires separately licensed vendor observations.

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Conflicts of interest

The author declares no conflicts of interest.

References

CME Clearing. (2020). *Valuation model change for four average-price options*. CME Clearing Advisory 20-198, May 15, 2020. https://www.cmegroup.com/notices/clearing/2020/05/Chadv20-198.pdf

CME Group. (2021). *NYMEX Rulebook Chapter 341: WTI Average Price Option*. NYMEX Submission No. 21-201, Exhibit A. https://www.cmegroup.com/market-regulation/rule-filings/2021/6/21-201.pdf

Curran, M. (1994). Valuing asian and portfolio options by conditioning on the geometric mean price. *Management Science*, *40*(12), 1705–1711.

Ewald, C.-O., Wu, Y., & Zhang, A. (2023). Pricing asian options with stochastic convenience yield and jumps. *Quantitative Finance*, *23*(4), 677–692. https://doi.org/10.1080/14697688.2022.2160799

Gilder, D., & Tsiaras, L. (2020). Volatility forecasts embedded in the prices of crude-oil options. *Journal of Futures Markets*, *40*(7), 1127–1159. https://doi.org/10.1002/fut.22114

Guidolin, M., & Timmermann, A. (2003). Option prices under bayesian learning: Implied volatility dynamics and predictive densities. *Journal of Economic Dynamics and Control*, *27*(5), 717–769. https://doi.org/10.1016/S0165-1889(01)00069-0

Kemna, A. G. Z., & Vorst, A. C. F. (1990). A pricing method for options based on average asset values. *Journal of Banking & Finance*, *14*(1), 113–129. https://doi.org/10.1016/0378-4266(90)90039-5

Matías González, A., Martínez-Palacios, M. T. V., & Ortiz-Ramírez, A. (2019). Consumo e inversión óptimos y valuación de opciones asiáticas en un entorno estocástico con fundamentos microeconómicos y simulación Monte Carlo. *Revista Mexicana de Economía y Finanzas, Nueva Época*, *14*(3), 397–414. https://doi.org/10.21919/remef.v14i3.408

Ortiz Ramírez, A., & Jiménez Preciado, A. L. (2026). Valuación de opciones asiáticas con volatilidad realizada y reversión a la media: Caso CEMEXCPO. *Revista Mexicana de Economía y Finanzas, Nueva Época*, *21*(1). https://doi.org/10.21919/remef.v21i1.1470

Rombouts, J. V. K., & Stentoft, L. (2014). Bayesian option pricing using mixed normal heteroskedasticity models. *Computational Statistics & Data Analysis*, *76*, 588–605. https://doi.org/10.1016/j.csda.2013.06.023

Sagor, H. H., Boone, E. L., & Ghanam, R. A. (2026). Bayesian joint estimation of the hurst parameter and volatility with applications to fractional option pricing. *Risks*, *14*(8). https://doi.org/10.3390/risks14080173

Schwartz, E. S. (1997). The stochastic behavior of commodity prices: Implications for valuation and hedging. *The Journal of Finance*, *52*(3), 923–973. https://doi.org/10.1111/j.1540-6261.1997.tb02721.x

Shiraya, K., & Takahashi, A. (2011). Pricing average options on commodities. *Journal of Futures Markets*, *31*(5), 407–439. https://doi.org/10.1002/fut.20481